

NLP TUTORIAL

LSTM with Attention

Content

- ✓ Introduction to Attention Mechanism and its need
- ✓ Attention based architecture
- ✓ Types of attention
- ✓ Self Attention
- ✓ Concept of query, key, value
- ✓ Calculation of attention score from query, key, value

Attention Mechanism



You want to recognize the peacock from this image

Think of a peacock in your mind , put more attention to the bird in the picture which matches with your imagination of a peacock.

Next word Prediction Problem

“Instead of paying through credit card, he started using UPI payment as it is more convinient.”

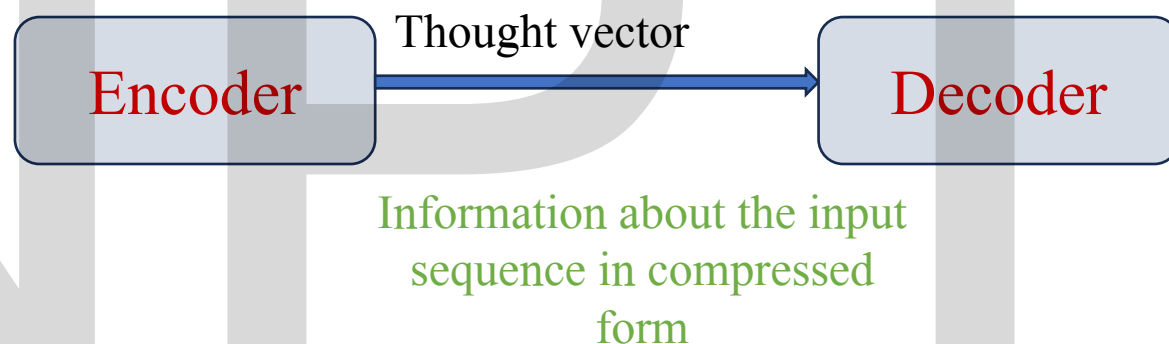
Instead of paying through credit card, he started using UPI payment as it is more

More attention

Basic LSTM gets confused between words and sometimes predict wrong words

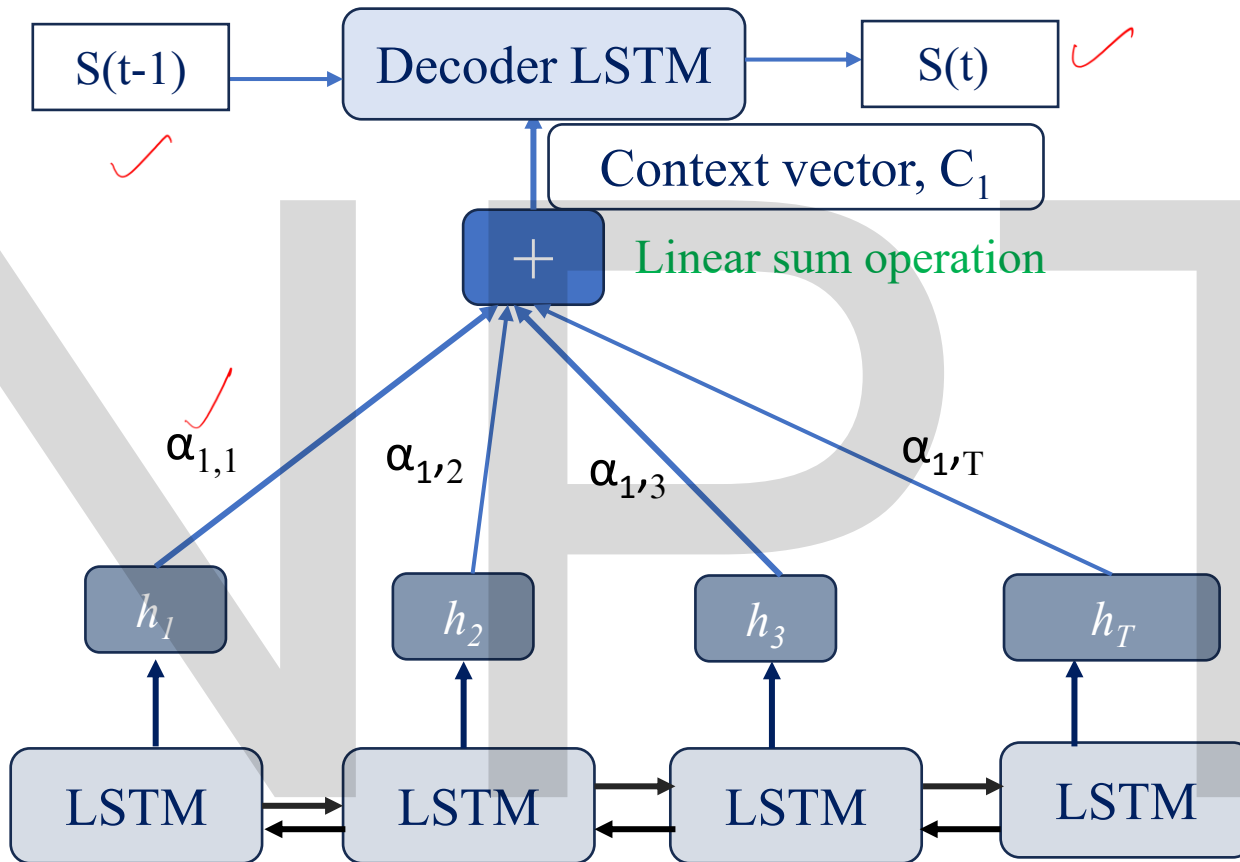
In simple Encoder Decoder architecture, decoder make predictions by looking only at the final output of the encoder which is a condensed information.

Difference between Vanilla Encoder-Decoder Architecture and Attention based Encoder Decoder architecture



Attention based architecture attends to every hidden state from encoder node at every time step and then make predictions after deciding which part is more important.

Attention based Architecture

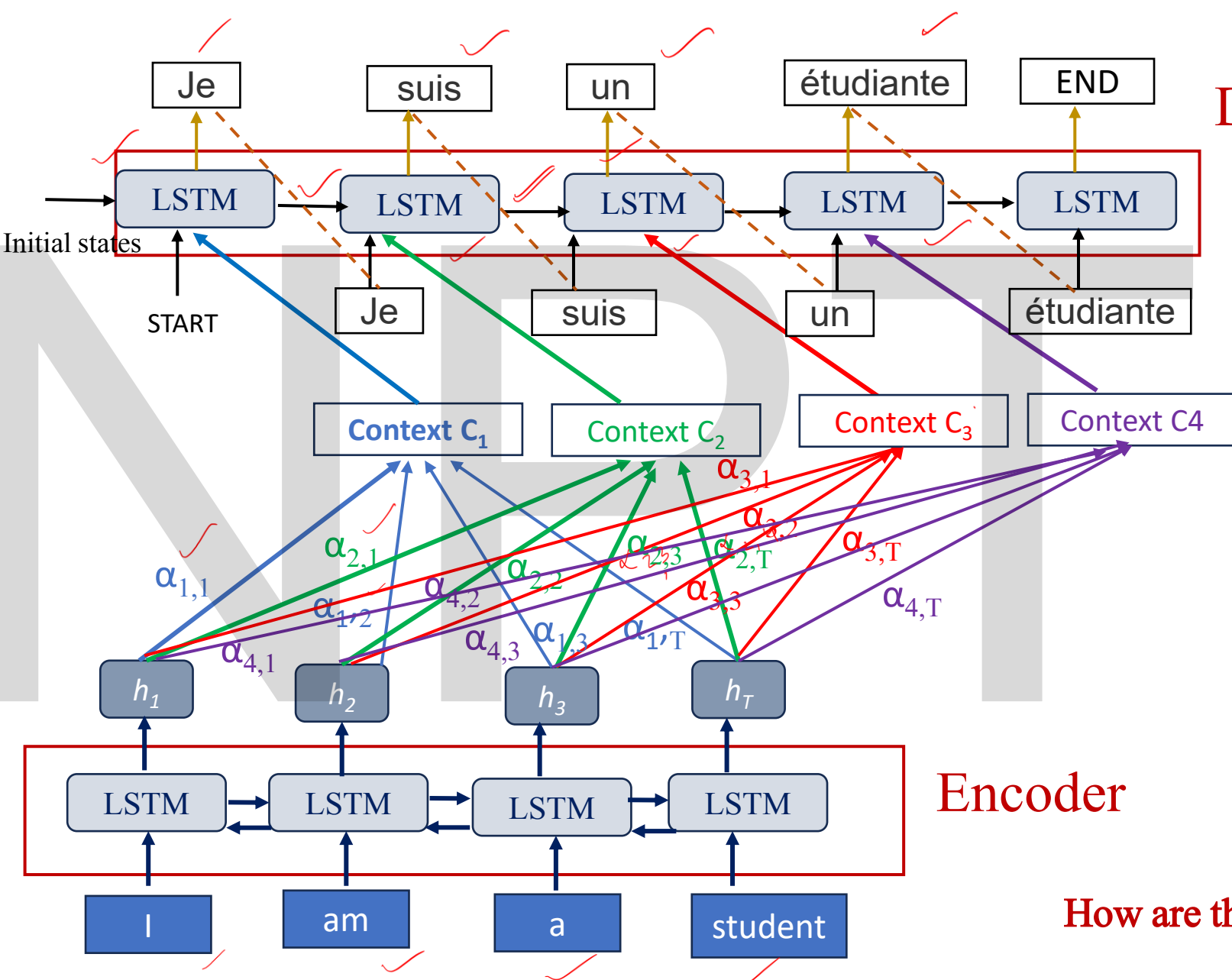


$$C_1 = \sum_{j=1}^T \alpha_{1,j} h_j$$

α denotes the amount of attention to be paid to each hidden state

The context vector is a linear combination of hidden state value h_j weighted by the attention value $\alpha_{t,j}$

$S(t-1)$ = Previous hidden state of decoder



Decoder

$$C_1 = \sum_{j=1}^T \omega_{1,j} h_j$$

$$C_2 = \sum_{j=1}^T \omega_{2,j} h_j$$

$$C_3 = \sum_{j=1}^T \omega_{3,j} h_j$$

$$C_4 = \sum_{j=1}^T \omega_{4,j} h_j$$

Encoder

How are the attention scores calculated?

Types of Attention Mechanism

Encoder Decoder Attention

Attention between the input sequence and the output sequence

Self Attention in input sequence

Attends to all words in the input sequence

Self Attention in output sequence

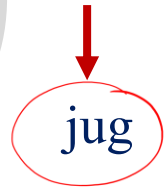
Attends to all words in the output sequence

Self Attention

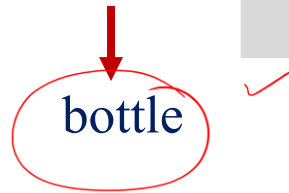
Self attention helps to create similar connections within the same sentence.

Ex:

I poured milk from the bottle into the jug till it's full



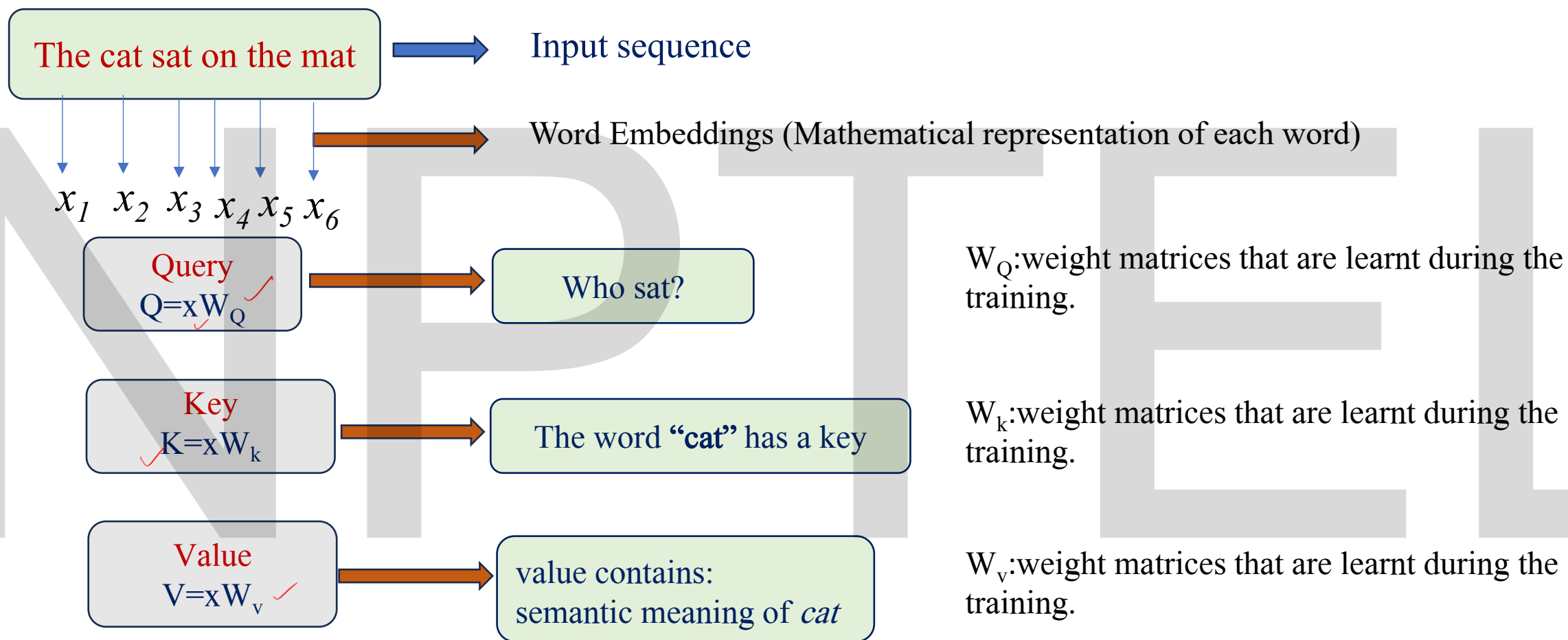
I poured milk from the bottle into the jug until it's empty



Self-attention answers one question:

“For a particular word (or token), which other words should I pay attention to, and how much?”

Key, Value and Queries



Each word now has:

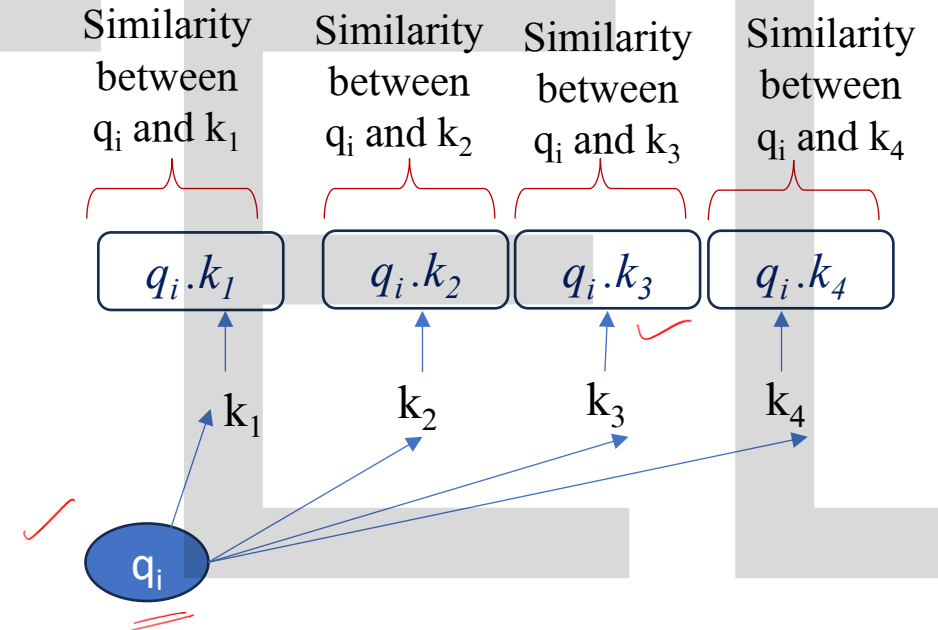
- ✓ a query vector
- ✓ a key vector
- ✓ a value vector

Calculation of Self Attention

Step 1:

- ✓ Finding the most similar query and key pair
- ✓ Done using cosine similarity
- ✓ Take the dot product of q_i and K_j
- ✓ Closest query key value will have the highest score

$$q_i \cdot k_j \quad \text{for all } 0 \leq j \leq n$$



Step 2:

- ✓ Divide the product by the dimension of the key vector
- ✓ Required for better gradient flow
- ✓ Important for the cases where the value of the dot product in the previous step is too big

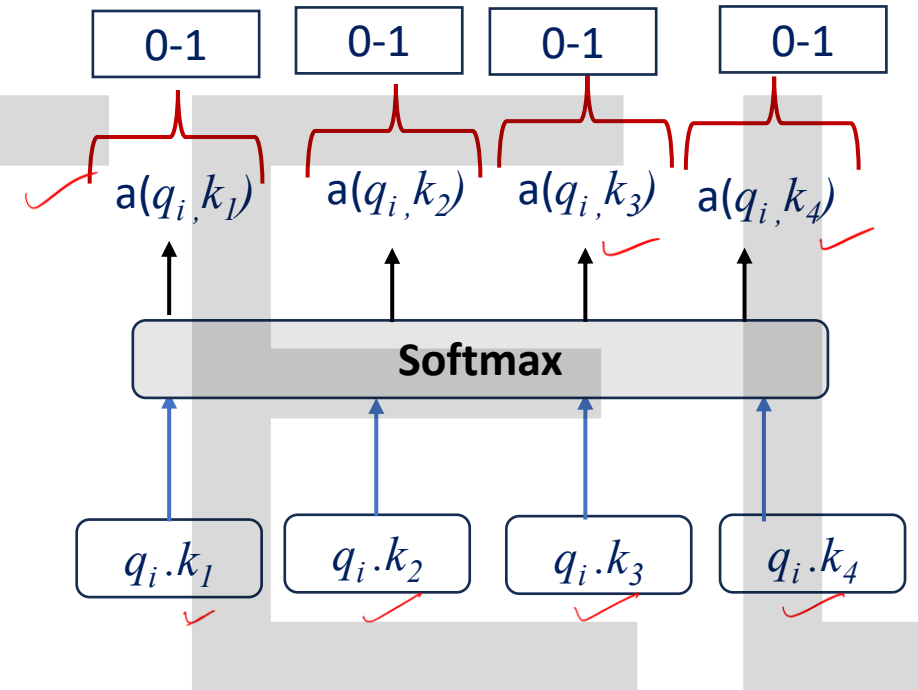
$$\frac{q \cdot k_j}{\sqrt{\dim(k_j)}} \quad \text{for all } 0 \leq j \leq n$$

Step 3:

- ✓ Once scores for all j 's are calculated, then they are passed through the softmax.
- ✓ Gives the normalized value of the scores.

$$\text{Softmax} \left[\frac{q \cdot k_j}{\sqrt{\dim(k_j)}} \right] \quad \text{for all } 0 \leq j \leq n$$

Softmax converts scores into **attention weights** (sum to 1)

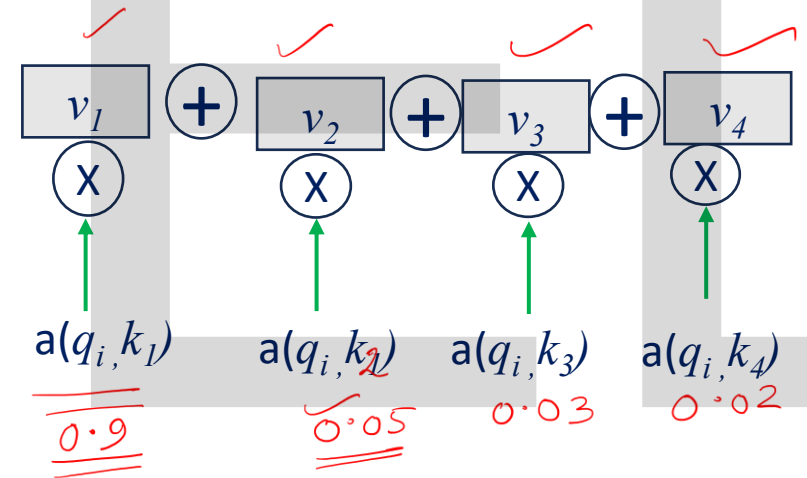


Step 4:

- ✓ Multiply the softmax score for each j with vector v_i .
- ✓ This preserves only the values(v) of the input words we want to focus on by multiplying them with high probability score from softmax.
- ✓ Removes the remaining by driving them towards zero.

$$\text{Softmax} \left[\frac{q \cdot k_j}{\sqrt{\dim(k_j)}} \right] \cdot v_i \quad \text{for all } 0 \leq j \leq n$$

$$Z_i = \sum_{j=0}^n \text{Softmax} \left[\frac{q \cdot k_j}{\sqrt{\dim(k_j)}} \right] \cdot v_i$$



Self Attention in input sequence

Query, Key, and Value matrices are all derived from the same input sequence, specifically the output of the previous encoder layer.

Self Attention in output sequence

Query, Key, and Value matrices are all derived from the output of the previous decoder layer.

Encoder Decoder Attention

Query matrix comes from the output of the previous decoder layer.
The Key and Value matrices come from the output of the encoder

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Summary

Next Session

Hands-on on building LSTM models for structured and unstructured data

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Thank you